(4) The algebra $\mathrm{aff}(1)$ of matrices $\left(\begin{smallmatrix} * & * \\ 0 & 0 \end{smallmatrix}\right)$.
Its basis consists of $X = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $Y = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, with $[X, Y] = Y$.

(5) $\mathfrak{so}(n)$, the space of skew-symmetric $n \times n$ matrices, with $[a, b] = ab - ba$.

Exercise 2.9.5. Show that example (1) is a special case of example (5) (for $n = 3$).

Definition 2.9.6. Let $\mathfrak{g}_1, \mathfrak{g}_2$ be Lie algebras. A **homomorphism of Lie algebras** $\varphi : \mathfrak{g}_1 \longrightarrow \mathfrak{g}_2$ is a linear map such that $\varphi([a, b]) = [\varphi(a), \varphi(b)]$ for all $a, b \in \mathfrak{g}_1$.

Definition 2.9.7. A **representation** of a Lie algebra $\mathfrak{g}$ is a vector space V with a homomorphism of Lie algebras $\rho : \mathfrak{g} \longrightarrow \mathrm{End}\, V$.

Example 2.9.8. Some examples of representations of Lie algebras are:

(1) $V = 0$.

(2) Any vector space V with $\rho = 0$ (the trivial representation).

(3) The adjoint representation $V = \mathfrak{g}$ with $\rho(a)(b) := [a, b]$.
That this is a representation follows from equation (2.9.1). Thus, the meaning of the Jacobi identity is that it is equivalent to the existence of the adjoint representation.

It turns out that a representation of a Lie algebra $\mathfrak{g}$ is the same thing as a representation of a certain associative algebra $\mathcal{U}(\mathfrak{g})$. Thus, as with quivers, we can view the theory of representations of Lie algebras as a part of the theory of representations of associative algebras.

Definition 2.9.9. Let $\mathfrak{g}$ be a Lie algebra with basis x_i and $[\ ,\ ]$ defined by $[x_i, x_j] = \sum_k c_{ij}^k x_k$. The **universal enveloping algebra** $\mathcal{U}(\mathfrak{g})$ is the associative algebra generated by the x_i's with the defining relations $x_i x_j - x_j x_i = \sum_k c_{ij}^k x_k$.

Remark 2.9.10. This is not a very good definition since it depends on the choice of a basis. Later we will give an equivalent definition which will be basis-independent.